

Homework #2

1. McIntyre, Chapter 1, Problem 1.16

2. A quantum object whose state is given by $|\psi\rangle = \frac{1}{2}|+\rangle_z - \frac{\sqrt{3}}{2}|-\rangle_z$ is sent through a Stern-Gerlach device with the magnetic field oriented in the y -direction. What is the probability that this object will emerge from the $+$ side of this device?

3. McIntyre, Chapter 2, Problem 2.3

4. Suppose that operators $\hat{A}$ and $\hat{B}$ are both Hermitian, i.e., $\hat{A}^\dagger = \hat{A}$ and $\hat{B}^\dagger = \hat{B}$. Answer the following and show your work:

(a) Is $\hat{A}^2$ Hermitian?

(b) Is $\hat{A}\hat{B}$ Hermitian?

(c) Is $\hat{A}\hat{B} + \hat{B}\hat{A}$ Hermitian?

(d) Is it possible for $\hat{A}$ to have complex eigenvalues, or must they be real?

5. Complete the next page of questions about Dirac Notation.

Change of basis

Suppose an electron is described by the state vector $|\psi\rangle = \frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle$, where $|+\rangle$ and $|-\rangle$ are the basis states corresponding to spin-up and spin-down, respectively, in the z -direction.

A. Calculate the inner product $\langle +|\psi\rangle$.

$$\begin{aligned}\langle +|\psi\rangle &= \langle +|\left[\frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle\right] = \frac{1}{2}\langle +|+\rangle + \frac{\sqrt{3}}{2}\langle +|-\rangle \\ &= \frac{1}{2}(1) + \frac{\sqrt{3}}{2}(0) = \frac{1}{2}\end{aligned}$$

B. Is this state vector normalized? Explain by discussing the probabilities of measuring spin up or down in the z -direction, as done in the Worksheet 2.

Yes, because $|\psi\rangle$ is normalized if its probability amplitudes (a^2 and b^2 in $|\psi\rangle = a|+\rangle + b|-\rangle$) sum to 1. In this case, $a = \frac{1}{2}$ and $b = \frac{\sqrt{3}}{2}$.
 $a^2 = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$ and $b^2 = \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$ $\frac{1}{4} + \frac{3}{4} = a^2 + b^2 = 1$

C. Predict (without performing explicit calculations) whether the magnitude of the inner product ${}_x\langle +|\psi\rangle$ is greater than, less than, or equal to the magnitude of the inner product $\langle +|\psi\rangle$, where $|+\rangle_x$ is the basis state corresponding to spin-up in the x -direction. Explain.

Greater than: finding $|+\rangle_x$ has extra nonzero coefficients and a $|-\rangle$ component so there'll be an extra additive term when finding $\langle +_x|\psi\rangle$ that makes the inner product larger than $\langle +|\psi\rangle$ alone.

D. Consider the dialogue below between three students who are attempting to determine the inner product ${}_x\langle +|\psi\rangle$.

Student 1: "This is just like the first inner product: ${}_x\langle +|-\rangle$ is zero because the plus and minus states are always orthogonal to each other. Therefore, ${}_x\langle +|\psi\rangle$ is $1/2$."

Student 2: "The x - and z -directions are orthogonal to each other, so $|+\rangle_x$ is orthogonal to both $|+\rangle$ and $|-\rangle$. That means the answer is zero."

Student 3: "I don't think the inner product of ${}_x\langle +|+\rangle$ is zero. If we send an electron prepared in the state $|+\rangle$ through a S-G apparatus with magnetic field in the x -direction, we would have a 50/50 chance of measuring spin up and spin down. Thus, the inner product can't be zero."

Which student(s), if any, do you agree with? Explain your reasoning.

Student 3: none of their discussion is related to $\langle +_x|\psi\rangle$ but they have the right logic. Probability is the squared absolute magnitude of the inner product, so to get a nonzero probability, $\langle +_x|\psi\rangle$ has to be nonzero. Student 1 doesn't account for the change in direction ($\langle +_x| \neq \langle +|$) and student 2 is right that $\langle +_x|$ is orthogonal to $\langle +|$ and $\langle -|$, but not on the calculations: coefficients and multiplying components!!

Akhila Pisumati

- E. Determine the inner product, ${}_x\langle +|\psi\rangle$, using Dirac notation. Clearly show each step in your calculation.

$$\begin{aligned}
 |+_x\rangle &= \frac{1}{\sqrt{2}}|+\rangle + \frac{1}{\sqrt{2}}|-\rangle, \text{ so } {}_x\langle +| = \frac{1}{\sqrt{2}}\langle +| + \frac{1}{\sqrt{2}}\langle -| \\
 {}_x\langle +|\psi\rangle &= \left[\frac{1}{\sqrt{2}}\langle +| + \frac{1}{\sqrt{2}}\langle -| \right] \left[\frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle \right] \\
 &= \frac{1}{2\sqrt{2}}\langle +|+\rangle + \frac{\sqrt{3}}{2\sqrt{2}}\langle +|-\rangle + \frac{1}{2\sqrt{2}}\langle -|+\rangle + \frac{\sqrt{3}}{2\sqrt{2}}\langle -|-\rangle \\
 &= \frac{1+\sqrt{3}}{2\sqrt{2}} \approx 0.97 > \frac{1}{2} = \langle +|\psi\rangle
 \end{aligned}$$

Resolve any inconsistencies with your prediction in question C.

- F. Is it possible to rewrite $|\psi\rangle$ in terms of the basis state vectors $|+_x\rangle$ and $|-_x\rangle$? Explain why or why not.

Yes, because you can write the basis vectors of $|\psi\rangle$, $|+\rangle$ and $|-\rangle$, entirely with respect to $|+_x\rangle$ and $|-_x\rangle$, not the other current basis vector. It stands to reason that some multiplication would yield $|\psi\rangle$ in terms only of $|+_x\rangle$ and $|-_x\rangle$!

If it is possible, do you expect the coefficients in terms of $|+_x\rangle$ and $|-_x\rangle$ to be the same as the coefficients in terms of $|+\rangle$ and $|-\rangle$? Explain.

No, because the coefficients represent projections of $|\psi\rangle$ along each respective basis ket. The state is not equally projected along the 'x' and 'z' directions, or their probabilities would be equal, which isn't happening here: $|\langle +|\psi\rangle|^2 = \frac{1}{4} \neq |\langle +_x|\psi\rangle|^2 = 0.97^2$

If it is possible to write $|\psi\rangle$ in terms of $|+_x\rangle$ and $|-_x\rangle$, do so. Clearly show your work.

$$|+\rangle = \frac{1}{\sqrt{2}}|+_x\rangle + \frac{1}{\sqrt{2}}|-_x\rangle$$

$$|-\rangle = \frac{1}{\sqrt{2}}|+_x\rangle - \frac{1}{\sqrt{2}}|-_x\rangle$$

$$|\psi\rangle = \frac{1}{2}|+\rangle + \frac{\sqrt{3}}{2}|-\rangle = \frac{1}{2\sqrt{2}}|+_x\rangle + \frac{1}{2\sqrt{2}}|-_x\rangle + \frac{\sqrt{3}}{2\sqrt{2}}|+_x\rangle - \frac{\sqrt{3}}{2\sqrt{2}}|-_x\rangle$$

$$\left\{ \begin{aligned} \frac{1}{2}|+\rangle &= \frac{1}{2} \left[\frac{1}{\sqrt{2}}(|+_x\rangle + |-_x\rangle) \right] = \frac{1}{2\sqrt{2}}|+_x\rangle + \frac{1}{2\sqrt{2}}|-_x\rangle \\ \frac{\sqrt{3}}{2}|-\rangle &= \frac{\sqrt{3}}{2} \left[\frac{1}{\sqrt{2}}(|+_x\rangle - |-_x\rangle) \right] = \frac{\sqrt{3}}{2\sqrt{2}}|+_x\rangle - \frac{\sqrt{3}}{2\sqrt{2}}|-_x\rangle \end{aligned} \right.$$

$$|\psi\rangle = \frac{1+\sqrt{3}}{2\sqrt{2}}|+_x\rangle + \frac{1-\sqrt{3}}{2\sqrt{2}}|-_x\rangle$$